Traficom Space & Terrestrial RF Interference Screening Engine

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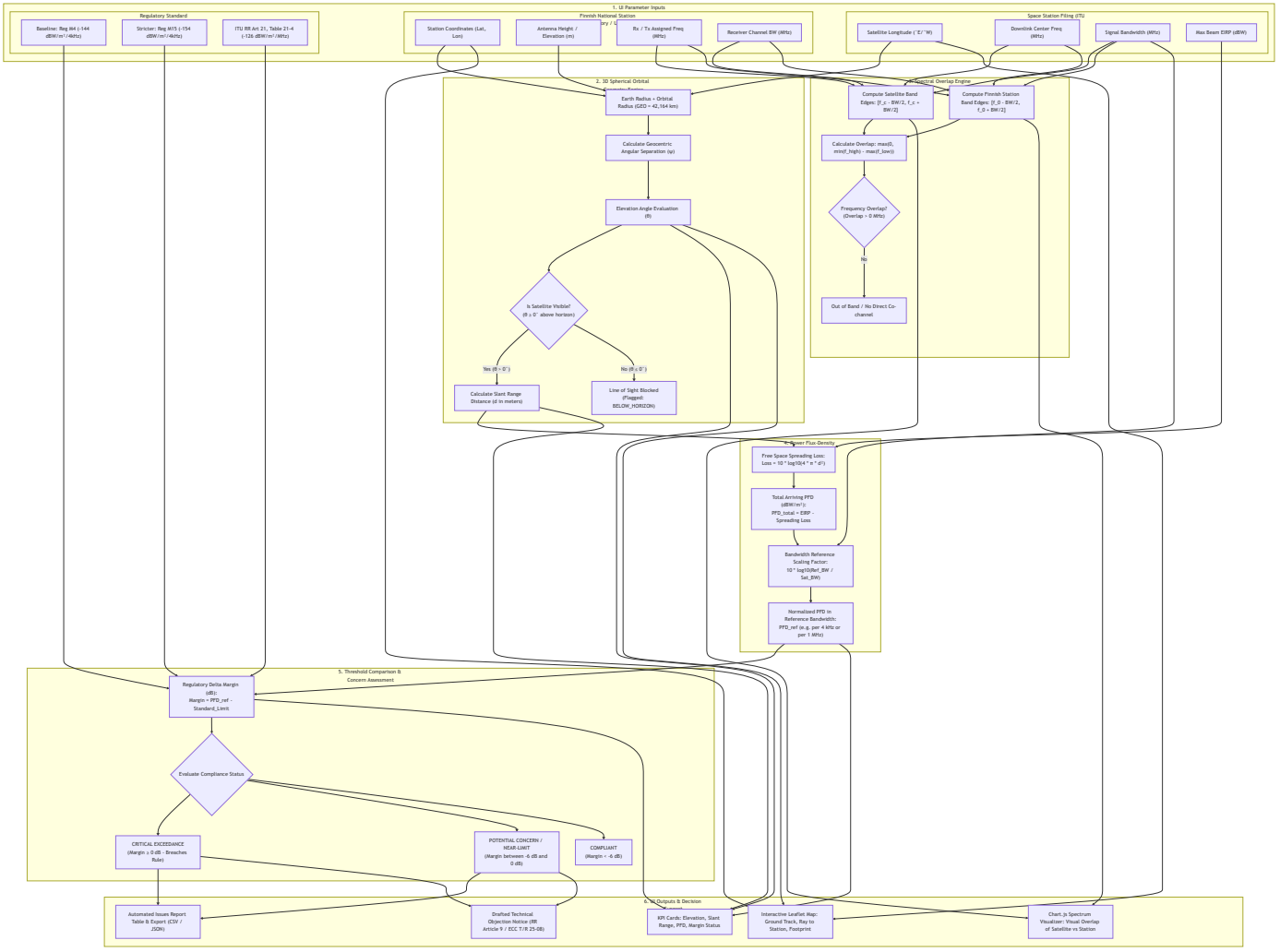
Technical Architecture, UI Element Dataflow & Customer Question Guide

This document breaks down the **Interactive Screening Engine** and its automated sub-systems into visual process flowcharts, mathematical explanations of UI elements, and a targeted questionnaire for Traficom spectrum managers.

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1. End-to-End Processing Flowchart

The diagram below maps every input field in the UI to its internal physics calculation, regulatory evaluation, and final visual outputs.



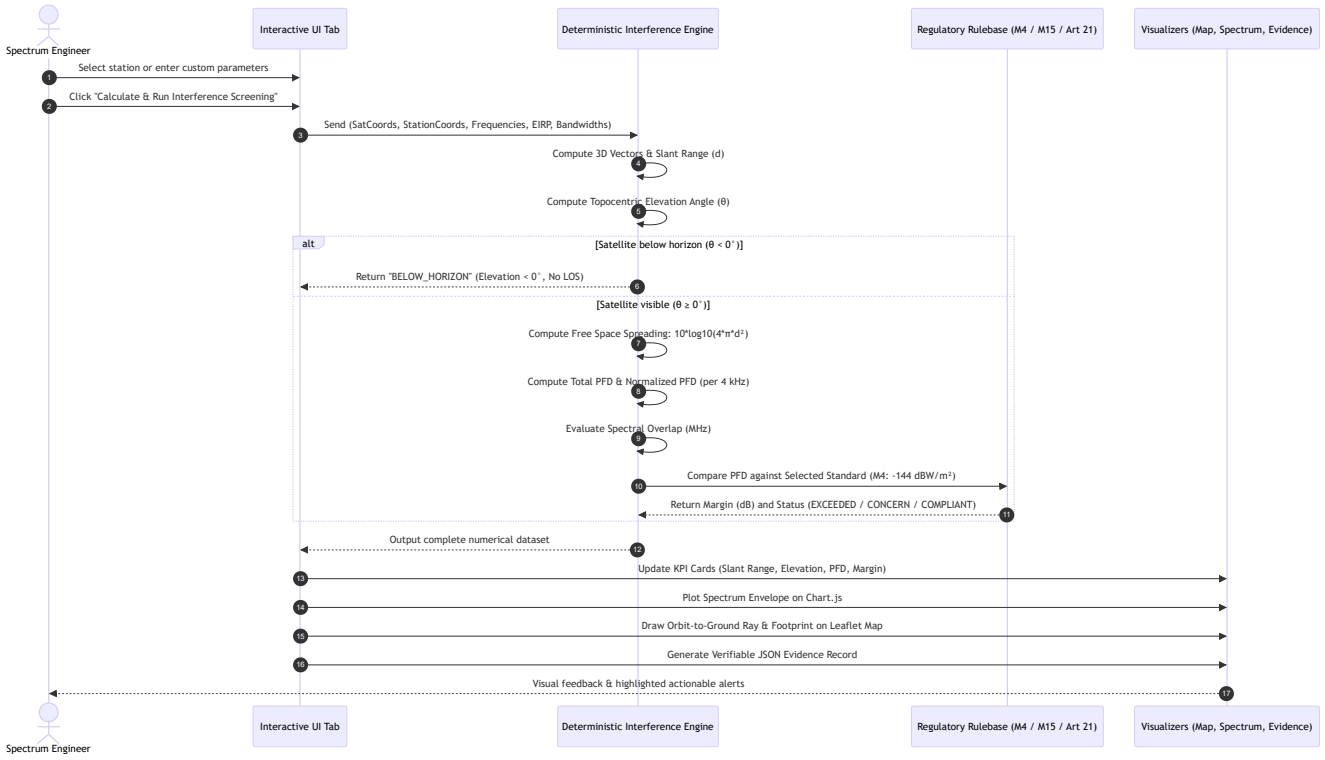
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2. Element-by-Element Breakdown

UI Element	Source / Reference	What Happens Internally	Why It Matters to Traficom
Satellite Orbital Longitude	ITU BR IFIC circulars (e.g. `25.0°E` or `13.0°E`)	Converts satellite orbital position on the equator into 3D ECEF Cartesian coordinates (X_s, Y_s, Z_s) at $R = 42,164\ \text{km}$.	Determines whether the satellite is physically above Finland's horizon ($\theta > 0^\circ$). If below, no line-of-sight path exists.
Finnish Station Coordinates & Ground Height	874-station Excel dataset (`Asemanro`, `Korkeus`)	Converts WGS84 latitude, longitude, and elevation into 3D station position (X_t, Y_t, Z_t).	Precise ground location dictates elevation angle and geometric distance to the geostationary arc.
Spectral Overlap Calculator	Downlink frequency & Channel bandwidth vs. Finnish Rx/Tx frequency	Evaluates $[\max(f_{s,\min}, f_{t,\min}), \min(f_{s,\max}, f_{t,\max})]$. If difference > 0, an overlap exists. Checks both Rx and Tx per the Traficom rule *"Every TX is received in Finland"*.	Identifies co-channel or overlapping interference where satellite energy enters the receiver passband.
Slant Range (d) & Free-Space Spreading	3D Euclidean distance vector	Calculates $d = \sqrt{\Delta X^2 + \Delta Y^2 + \Delta Z^2}$ (typically $38,000 - 41,000\ \text{km}$ at Finnish latitudes). Spreading attenuation is $10\log_{10}(4\pi d^2) \approx 162.5 - 163.3\ \text{dB}$.	Standard inverse-square energy divergence across interplanetary distance.
Power Flux-Density (PFD) Scaling	Sat EIRP (dBW) & Bandwidth	Arriving $\text{PFD}_{\text{total}} = \text{EIRP} - 10\log_{10}(4\pi d^2)$. Scaled to reference bandwidth ($4\ \text{kHz}$ or $1\ \text{MHz}$) using $10\log_{10}(\text{BW}_{\text{ref}}/\text{BW}_{\text{sat}})$.	Satellite filings declare peak EIRP across wide carriers (e.g. $20\ \text{MHz}$ or $36\ \text{MHz}$); limits are normalized per $4\ \text{kHz}$ slice.
Regulatory Evaluation Standard	Dropdown: Regulation M4, M15,	Margin $\Delta = \text{PFD}_{\text{calc}} - \text{PFD}_{\text{limit}}$. If $\Delta \geq 0\ \text{dB}$, flagged **CRITICAL EXCEEDANCE**. If	Automatically highlights filings requiring

UI Element	Source / Reference	What Happens Internally	Why It Matters to Traficom
	ITU RR Art 21	$-6\ \mathrm{dB} \leq \Delta < 0\ \mathrm{dB}$, flagged **POTENTIAL CONCERN / NEAR-LIMIT**.	formal objection within the ITU 4-month regulatory window.

3. Sequence Diagram: What Happens When You Click "Run Screening"



4. Key Questions to Ask the Customer (Traficom Specialists)

Use these targeted questions in interviews or customer alignment sessions:

A. Regulatory Thresholds & Margins

1. **M4 vs. M15 Baseline:** "We have configured **Regulation M4 ($-144\ \mathrm{dBW/m^2/4kHz}$)** as the default baseline, with **Regulation M15 ($-154\ \mathrm{dBW/m^2}$)** selectable for sensitive mobile/terrestrial systems. Are there specific frequency ranges where M15 should be enforced automatically rather than by manual selection?"
2. **Concern Threshold Window:** "Our engine flags any filing within $6\ \mathrm{dB}$ **of the limit** as a `POTENTIAL CONCERN` even if it does not strictly exceed the hard limit. Does Traficom prioritize bilateral coordination inquiries for near-limit cases within $3\ \mathrm{dB}$, $6\ \mathrm{dB}$, or a wider margin?"

B. Antenna Directivity & Terrain Shielding

3. **Worst-Case Peak Envelope vs. Off-Axis Discrimination:** "Per official advice, we evaluate the worst-case peak beam EIRP envelope because boresight coordinates are often omitted in early IFIC filings. When coordination advances, would you want the tool to ingest ITU-R S.465/S.580 earth station antenna patterns to model off-axis rejection?"
4. **Digital Elevation Models (DEM):** "For terrestrial microwave fixed links, does Traficom require terrain profile obstruction checks (SRTM 30m) along the path, or is geometric line-of-sight sufficient for the initial screening stage?"

C. Terrestrial Cross-Border (ECC T/R 25-08)

5. **Border Coordination Threshold:** "For the land-mobile side quest, we calculate field strength E along 22 points across the Finnish borders. Is the standard $30\ \mathrm{dB}\mu\mathrm{V}/\mathrm{m}$ at 3m antenna height appropriate for all bands, or should we incorporate specific thresholds for 700 MHz (Band 28) and 3.5 GHz (Band n78)?"
6. **Sea vs. Land Propagation Weights:** "Across the Gulf of Finland from Tallinn, ducting and low sea clutter maintain high field strengths over 80 km. Do you prefer ITU-R P.1546 50% time / 50% locations or 10% time (worst-case anomalous propagation) for cross-border coordination?"